SETS IN PROBABILITY

The set notation of probability

Symbols from set theory, each with the meaning it takes inside a sample space.

01 Listing and membership

3 symbols

$\{\ \}$ **Set braces**
lists the outcomes inside a set

$\in$ **Element of**
an outcome belongs to a set

$\notin$ **Not an element of**
an outcome does not belong to a set

02 Containment

2 symbols

$\subseteq$ **Subset**
one set is contained inside another

$\subset$ **Strict subset**
contained but not equal

03 Combining events

4 symbols

$\cup$ **Union**
either event occurs

$\cap$ **Intersection**
both events occur

$\setminus$ **Difference**
in one event but not the other

A^c **Complement**
the complement of event A, also written A'

04 Special sets

2 symbols

$\varnothing$ **Empty set**
contains no outcomes

Ω **Sample space**
the full set of all possible outcomes

05 Intervals and size

2 symbols

$[a, b]$ **Intervals**
used for continuous outcomes: (a, b), $[a, b]$, $(-\infty, x]$

$|A|$ **Size of a set**
the number of elements in a finite set A